

Name of research:

Dynamic 1:k Matching Analysis of Preventive Mastectomy Effectiveness in Breast Cancer

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Motivation and background:

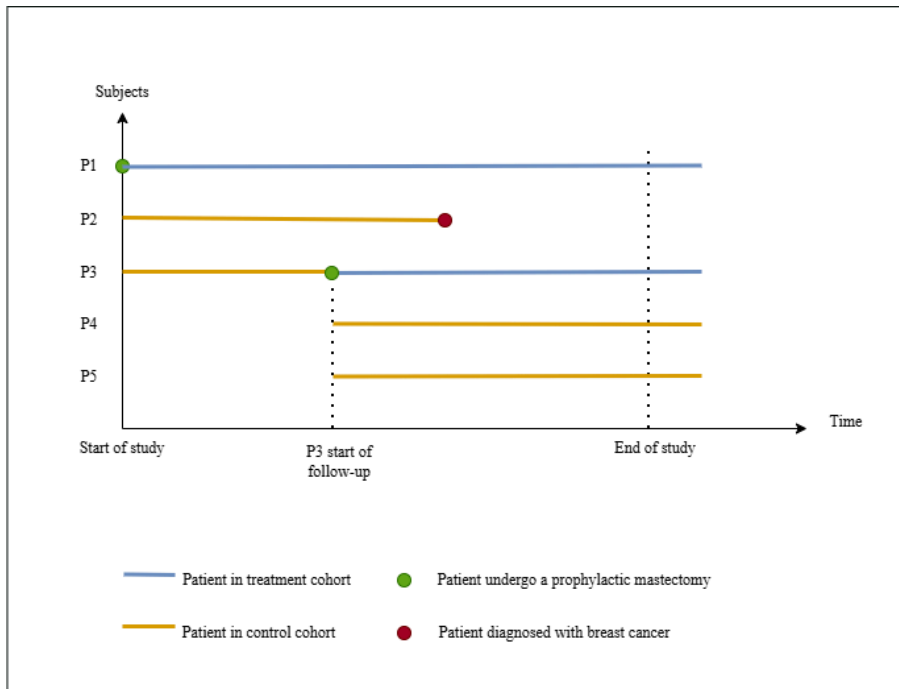
Breast cancer is the most common malignancy among women in Israel and Western countries, with a lifetime incidence of approximately one in eight Israeli women.[1] While the majority of cases (85-90%) are non-hereditary, certain genetic mutations are associated with significantly elevated risk. The BRCA1 and BRCA2 mutations, particularly prevalent among Ashkenazi, Ethiopian, and Iraqi populations, account for 50-80% of hereditary breast cancer cases and substantially increase lifetime disease susceptibility. Although complete prevention of breast cancer remains elusive, effective risk-reduction interventions exist, e.g., Prophylactic mastectomy, which is a surgical removal of breast tissue in high-risk individuals.[2,3]

This study will evaluate the long-term effectiveness of prophylactic mastectomy in high-risk patients using statistical matching. Matching is a technique that estimates treatment effects by pairing treated patients with comparable untreated individuals based on observable related characteristics, thereby constructing a comparison group.[4]

This analysis adapts the methodology of Dagan et al.[5] from Covid-19 vaccination effectiveness to mastectomy effectiveness, while implementing 1:k matching rather than 1:1 matching. Specifically, each treated patient is matched with multiple control patients, an approach anticipated to yield lower variance in the effect estimates. The matching is performed dynamically, allowing cohort composition to change throughout the follow-up period. At each assessment point, control group patients may undergo prophylactic mastectomy and consequently become ineligible as controls. Such patients are transferred to the treatment cohort and matched with k new control patients.[5]

This dynamic matching approach presents a methodological challenge: determining the appropriate follow-up endpoint when multiple matches exist per treated patient. Specifically, the analysis must address whether the follow-up period ends when one matched control undergoes mastectomy, or whether a minimum number of remaining matches is required.

Figure 1: Dynamic 1:2 matching demonstration



Problem definition:

While statistical matching methods exist for static cohorts with multiple controls and for dynamic $1:1$ matching, the combination of these approaches, dynamic $1:k$ matching, lacks established methodology in the literature. This creates several unresolved challenges.

The primary methodological gap concerns follow-up endpoint definition. When a treated patient is matched with k controls and one or more controls subsequently receive treatment, it remains unclear whether follow-up should terminate immediately, continue until a minimum number of matches remain, or follow another rule.

Additionally, appropriate estimators for this dynamic context must be determined. Traditional estimators for static cohorts may introduce bias when cohort composition changes over time, particularly with censored data. In this dynamic framework, censoring occurs when a control patient undergoes a mastectomy and transitions to the treatment group. Standard methods may not appropriately account for this dynamic matching process and may lead to incorrect conclusions about treatment effectiveness.

This project addresses these gaps by developing a comprehensive $1:k$ dynamic matching framework applicable to evaluating prophylactic mastectomy effectiveness.

Research stages:

1. **Literature review of estimation methods:** Examine existing estimators for simplified scenarios, including static cohorts with complete data and static cohorts with censored data.
2. **Performance metric specification:** Establish metrics for evaluating estimator quality to guide subsequent development.
3. **Estimator development:** Propose estimators tailored to dynamic cohorts with censored data.
4. **Covariate Selection:** Identify and select the covariates to best describe patient matching.
5. **Simulation and estimator evaluation:** Apply the proposed estimators to both synthesized data and real data from Clalit and Leumit health maintenance organizations. Assess estimator performance using the established metrics.

Note: The work with synthesized data will be conducted during the research project since the researcher doesn't have direct access to the real sensitive data. The final evaluation will be conducted on the real data with the help of the supervisor.

Possible extension: Derive theoretical properties of the proposed estimators

Literature Review:

This research examines statistical matching with multiple controls in censored data environments with dynamic cohorts. The literature review draws upon two use cases exemplifying key characteristics of the target environment, extracting methodologically sound approaches applicable to this study. The first is Dagan et al. (2021), which presents 1:1 matching with censored data and dynamic cohorts in assessing Covid-19 vaccination effectiveness [5]. The second is Yoo et al. (2025), demonstrating 1:4 matching with censored data examining the relationship between Parkinson's disease (PD) and benign paroxysmal positional vertigo (BPPV) [10]. Two critical decisions for successful research implementation are identified: matching paradigm selection and analysis method choice. This review evaluates approaches employed in both use cases and justifies selections appropriate for the current research.

The matching paradigm aims to approximate randomized controlled trial (RCT) conditions, making it essential for valid treatment-control comparisons [7]. When implementing multiple control matching, the bias-variance trade-off requires careful consideration. While increased control numbers reduce variance, additional controls may introduce bias, as they are necessarily more dissimilar from the treated patient than the closest match [7].

Yoo et al. employed 1:4 Fixed Matching, assigning exactly four controls per treated patient [10]. Generally, 1:k Fixed Matching assigns precisely k controls per treated patient. This approach typically utilizes optimal matching algorithms that minimize the total distance between treated and control participants across the dataset [6]. Distance is conventionally measured as the absolute difference in propensity scores (or their logits), representing each participant's probability of receiving treatment [6]. Dagan et al.'s 1:1 approach represents a special case of Fixed Matching where $k=1$. Due to its dynamic nature, matching occurred daily for newly treated participants, precluding global optimization in favor of a greedy matching algorithm. This sequential algorithm selects the most similar available control for each treated participant, often requiring similarity within a defined threshold (caliper). While Dagan et al. did not implement a caliper, it represents common practice in the literature [5,6,7]. An alternative not employed in these use cases is Varying Ratio Matching, which permits different control numbers per treated participant. Maximum control numbers may be specified but are not mandatory. Literature favors Varying Ratio Matching for minimizing bias, as Fixed Matching may necessitate dissimilar controls when the control pool is limited [7]. Varying Ratio Matching typically employs the greedy algorithm described above [7].

This research initially proposed Fixed Matching. However, given the advantages noted in the literature, Varying Ratio Matching warrants consideration or comparative evaluation. Regarding matching algorithms, Dagan et al.'s periodic greedy matching approach appears well-suited for dynamic cohort scenarios. For Fixed Matching, optimal algorithms are recommended, with the initial matching at the beginning of the trial employing optimal matching, followed by periodic greedy matching for subsequent additions. For Varying Ratio Matching, greedy algorithms are appropriate throughout.

Analysis method selection is crucial for accurate effect estimation [8]. Dagan et al. employed the Kaplan-Meier survival analysis estimator, appropriate for their 1:1 design [5]. However, Kaplan-Meier assumes observation independence, which holds for 1:1 matching but fails with multiple controls, as controls are selected based on covariate similarity to treated participants, creating dependence. Yoo et al. employed the Stratified Cox Proportional Hazards Model as an alternative [10]. This Cox regression variant estimates treatment effects on time-to-event outcomes for grouped or matched data. The Stratified Cox model offers two key advantages for this research. First, it accommodates varying baseline hazards between treatment-control sets, appropriate given matching based on propensity score distance naturally produces varying hazards across sets. The baseline hazard is defined as a nonnegative function of time, however, it factors out in the partial likelihood formula, eliminating the need for its explicit calculation [9]. Second, it handles censored data by accounting for varying set sizes and follow-up periods.

These characteristics also suit dynamic environments where follow-up periods naturally vary between sets [7,8,10]. An additional implementation advantage is that the Stratified Cox Proportional Hazards Model accommodates varying set sizes, making it suitable for both Fixed Matching and Varying Ratio Matching approaches.

Despite being the most appropriate method identified for this research context, the Stratified Cox Model has limitations that warrant acknowledgment. This approach yields conditional estimates of treatment effects rather than marginal estimates. Whereas randomized controlled trials typically report marginal hazard ratios characterizing population-level effects, conditional hazard ratios describe effects at the individual level. Furthermore, conditional estimates may result in amplified treatment effect magnitudes compared to marginal effects [8].

This review has synthesized matching paradigms and analysis methods from relevant literature and empirical research applicable to the current study's objectives. The methodological framework established provides a robust foundation for conducting rigorous causal inference in the presence of censored data and dynamic cohorts, balancing theoretical optimality with practical implementation considerations.

Results:

To evaluate the performance of dynamic 1:k matching under different design choices, eight matching experiments were conducted on the simulated cohort. The experiments crossed two factors: the assignment paradigm (fixed-ratio matching with optimal assignment vs. variable-ratio matching with greedy assignment) and the matching ratio ($K = 1, 2, 4, 8$ for each paradigm).

Prior to effect estimation, the quality of each matching configuration was evaluated by the reduction in propensity score distance between matched units relative to random pairing. Figure 2 reports the average propensity score distance between matched pairs before and after matching. Across all eight configurations, random pairing produced an average distance of approximately 0.054–0.057 on the logit scale, while matching reduced this distance by between one and two orders of magnitude. Fixed-ratio matching achieved near-perfect balance for $K=1$, $K=2$, and $K=4$ (all ≈ 0.000) and only a negligible residual distance at $K=8$ (0.001), reflecting the global optimality of the fixed-ratio assignment. Varying-ratio matching achieved comparable balance at the lowest ratio (max=1: 0.000) but degraded monotonically as the maximum ratio increased (max=2: 0.010; max=4: 0.013; max=8: 0.014). This pattern reflects the expected behavior of greedy variable-ratio assignment: as the algorithm is permitted to include more

controls per treated unit, it is forced to accept progressively less similar matches from the available pool.

Figure 2:

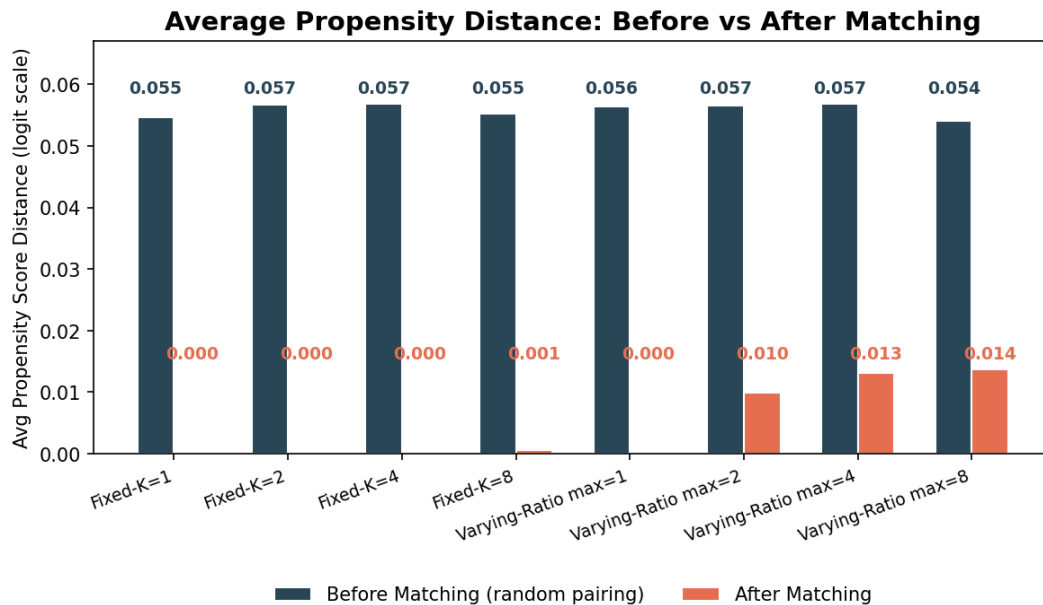


Figure 3 summarizes the estimated hazard ratios across all eight matching configurations alongside the ground-truth effect derived from the full simulated population.

Figure 3:

Matching Configuration	Risk Reduction (%)	Hazard Ratio (HR)	True HR – Estimated HR	SE	MSE	95% CI
Ground Truth	91.79	0.0821	-	-	-	-
Fixed K=1	91.45	0.0855	0.0034	0.0175	0.000317	[0.0573, 0.1277]
Fixed K=2	91.28	0.0872	0.0051	0.0152	0.000258	[0.0619, 0.1228]
Fixed K=4	90.23	0.0977	0.0157	0.0143	0.000451	[0.0733, 0.1303]
Fixed K=8	89.45	0.1055	0.0235	0.0141	0.000750	[0.0812, 0.1372]
Varying K=1	91.45	0.0855	0.0034	0.0175	0.000317	[0.0573, 0.1277]
Varying K=2	89.49	0.1051	0.0230	0.0173	0.000825	[0.0761, 0.1449]
Varying K=4	89.38	0.1062	0.0241	0.0153	0.000817	[0.0801, 0.1409]
Varying K=8	89.28	0.1072	0.0251	0.0144	0.000838	[0.0825, 0.1394]

All eight estimators recovered the true treatment effect with high accuracy, with absolute deviations from the ground-truth hazard ratio (HR = 0.0821) ranging from 0.0034 to 0.0251. The 95% confidence intervals of all eight estimators contained the true hazard ratio. The estimated reduction in breast cancer risk attributable to mastectomy ranged from 89.28% to 91.45%, consistent with prior clinical literature reporting risk reductions exceeding 90% [3]. At the lowest ratio, the two paradigms coincided (Fixed K=1 and Varying max=1 both yielded HR = 0.0855), as expected, since a maximum of one control per treated unit reduces variable-ratio matching to the 1:1 case.

A visual analysis of the standard error, bias, and mean squared error across all experiments is shown in Figures 4, 5, and 6, respectively:

Figure 4:

Coefficient Standard Error: Fixed vs Varying (Lower = More Precise)

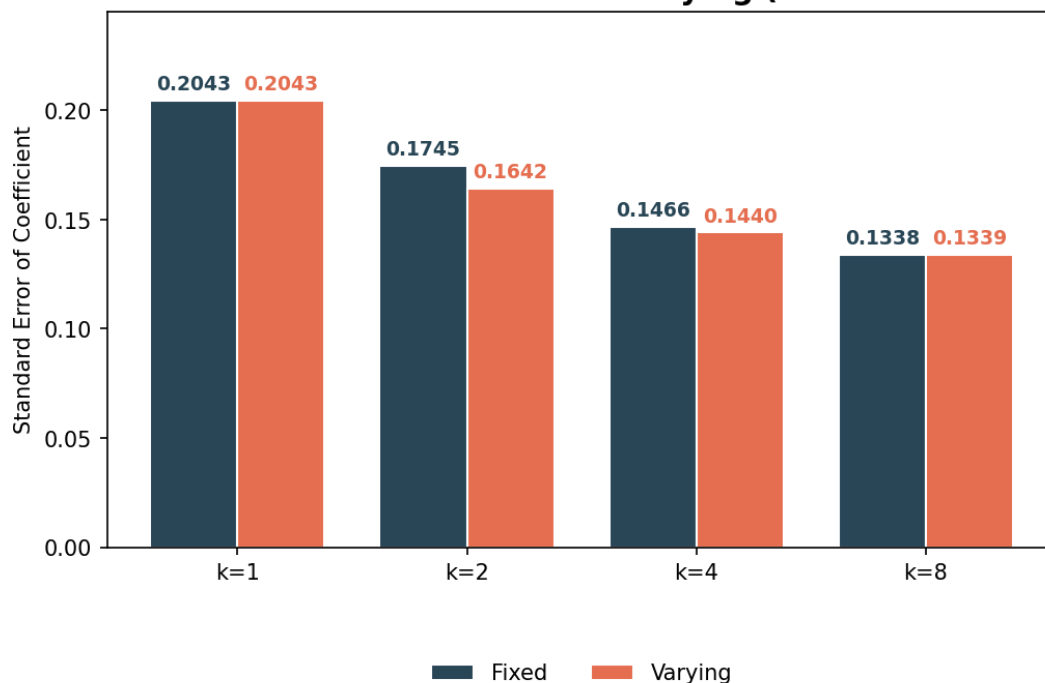


Figure 5:

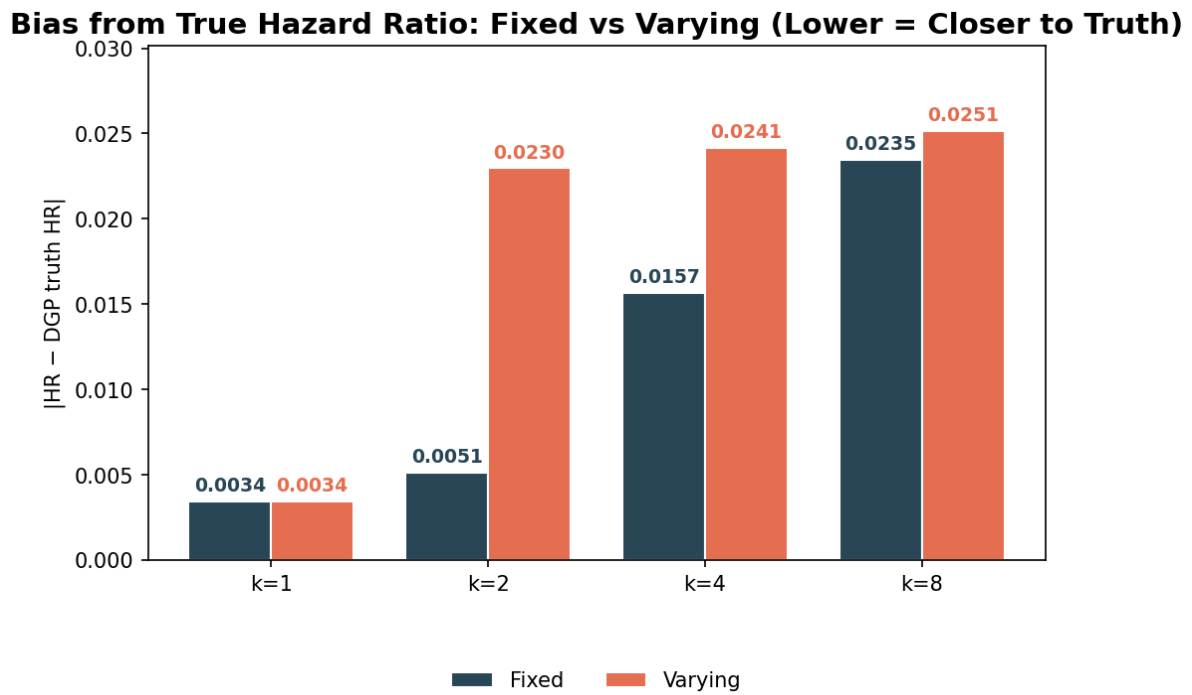
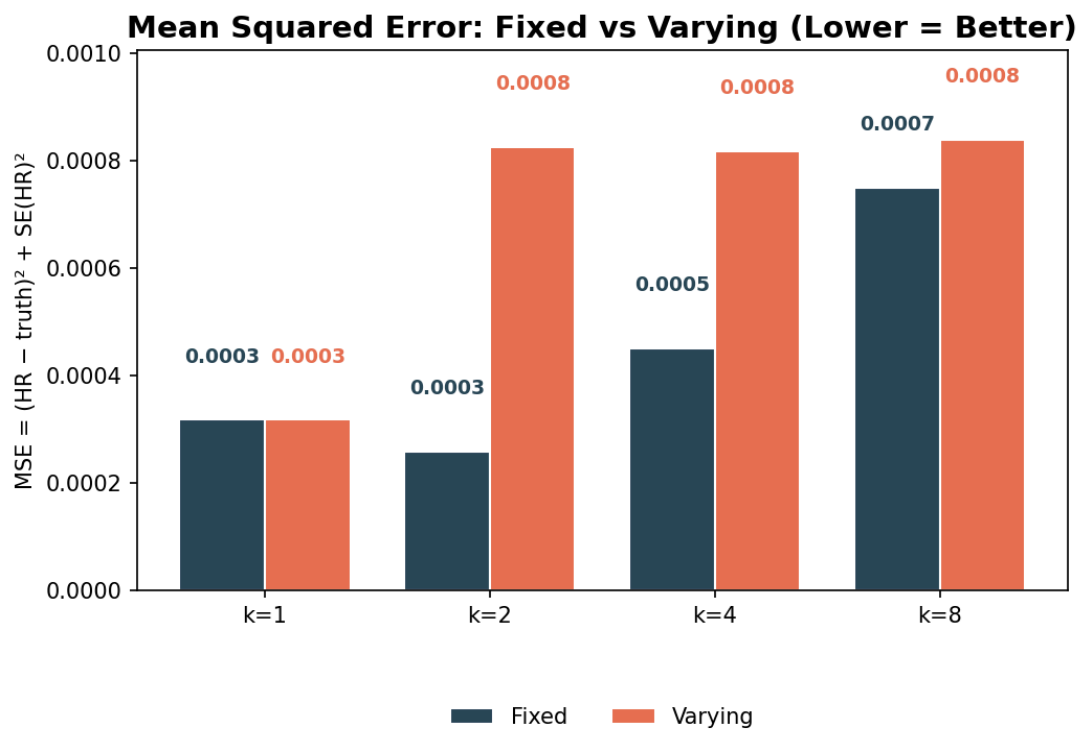


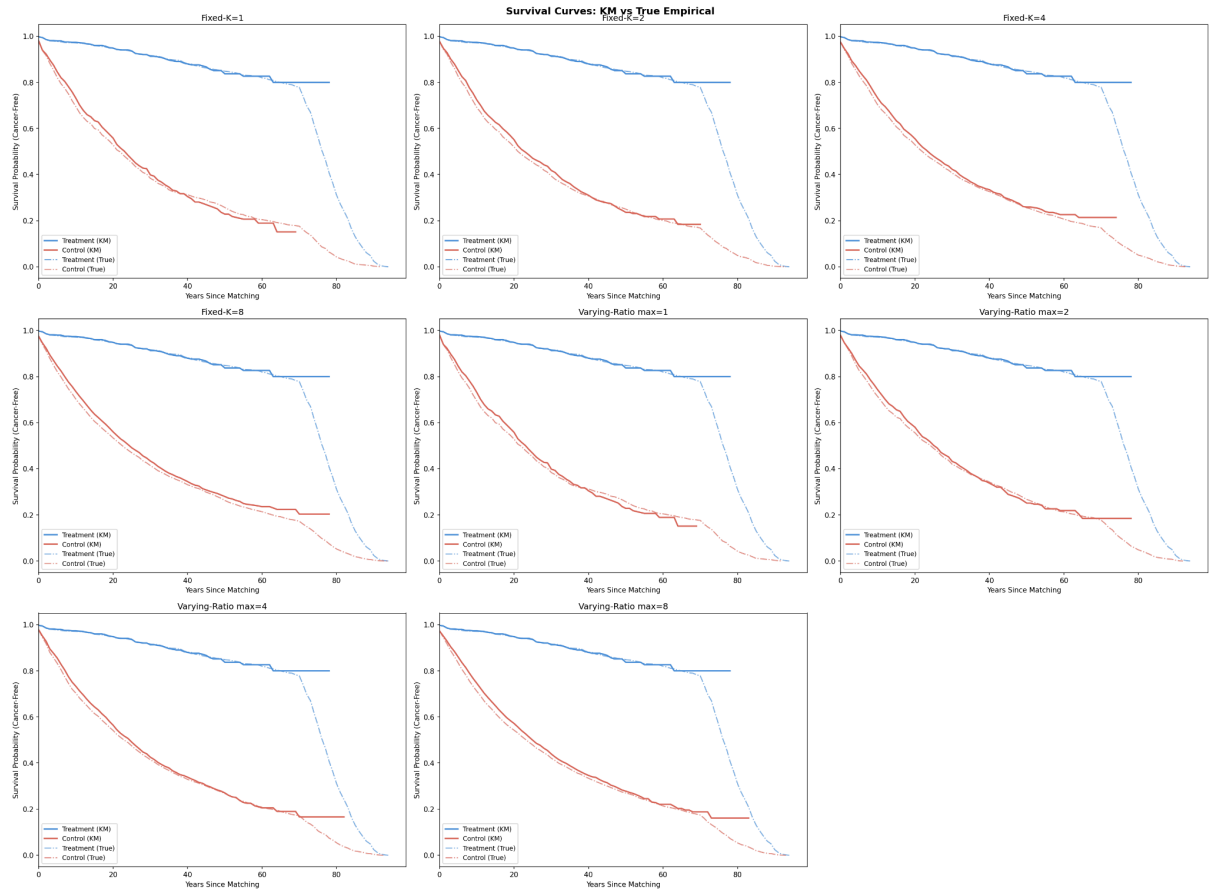
Figure 6:



The three figures expose a clear divergence between the precision and bias profiles of the two paradigms. Standard errors decreased monotonically with the matching ratio under both paradigms and were nearly indistinguishable between Fixed and Varying at every ratio (e.g., $k=2$: 0.1745 vs. 0.1642; $k=4$: 0.1466 vs. 0.1440; $k=8$: 0.1338 vs. 0.1339). In contrast, the bias profile differed sharply between paradigms. Fixed-ratio matching exhibited gradually growing bias as K increased ($0.0034 \rightarrow 0.0051 \rightarrow 0.0157 \rightarrow 0.0235$), while variable-ratio matching incurred a substantially larger bias penalty already at $\text{max}=2$ (0.0230) and remained elevated at higher ratios (0.0241 at $\text{max}=4$; 0.0251 at $\text{max}=8$). The mean squared error in Figure 6, defined as $\text{MSE} = (\text{HR} - \text{truth})^2 + \text{SE}(\text{HR})^2$, summarizes this bias–variance tradeoff in a single quantity. At the lowest ratio, the two paradigms coincided ($k=1$: 0.0003 for both), as expected. As the matching ratio increased, the two paradigms diverged sharply: under fixed-ratio matching, the MSE first decreased to its minimum at $K=2$ (0.0003) and then rose modestly at $K=4$ (0.0005) and $K=8$ (0.0007), reflecting a favorable balance between the precision gains from additional controls and the bias incurred by accepting less similar matches. Under variable-ratio matching, by contrast, the MSE jumped to 0.0008 already at $\text{max}=2$ and remained at that level through $\text{max}=4$ (0.0008) and $\text{max}=8$ (0.0008), as the bias penalty dominated any precision gains. Fixed-ratio matching therefore achieved a lower MSE than variable-ratio matching at every $K \geq 2$, with the gap most pronounced at $K=2$, where fixed-ratio matching reached its overall minimum.

Figure 7 presents the Kaplan–Meier survival curves for treated and control cohorts under each matching configuration, overlaid with the true empirical survival curves from the full underlying population. Across all eight configurations, the estimated curves closely track the true survival functions throughout follow-up, with divergence appearing only at late time points due to administrative censoring of the matched cohorts.

Figure 7:



The Kaplan–Meier estimator remains valid in this setting despite the within-stratum dependence induced by matching. While the same dependence motivated the use of a stratified Cox model for effect estimation (as discussed in the literature review), it does not compromise the Kaplan–Meier curves: because the estimator depends only on event and at-risk counts at each time point, $\hat{S}(t) = \prod_{t_i \leq t} (1 - d_i/n_i)$, the dependence among matched controls does not bias the point estimate, only its variance. Inference on the treatment effect is therefore drawn from the stratified Cox model, which appropriately accounts for the matched design, while the Kaplan–Meier curves serve a purely descriptive role.

Conclusions:

The risk reduction was estimated with reasonable accuracy across all eight matching configurations, with every 95% confidence interval containing the true hazard ratio. Increasing the matching ratio from $K=1$ to $K=8$ produced a substantial and monotonic reduction in standard error under both paradigms, and the two paradigms achieved essentially identical precision at every ratio. In this respect, the variance benefit of recruiting additional controls per treated unit was realized regardless of whether the additional controls were assigned through optimal fixed-ratio matching or greedy variable-ratio matching.

The two paradigms diverged sharply, however, in the accuracy of the resulting point estimates. Bias relative to the ground-truth hazard ratio grew with the matching ratio under both paradigms, but the rate of growth was markedly different: under fixed-ratio matching, the absolute deviation rose gradually from 0.0034 at $K=1$ to 0.0235 at $K=8$, whereas under variable-ratio matching it jumped to 0.0230 already at $\max=2$ and continued rising thereafter. This divergence is consistent with the propensity score balance results, which showed that fixed-ratio matching preserved near-perfect covariate balance up to $K=4$, while variable-ratio matching incurred residual imbalance at every ratio above $\max=1$. The pattern reflects a fundamental tradeoff inherent to ratio matching: requiring more controls per treated unit forces the inclusion of less similar pairs, and a greedy assignment rule incurs this cost more aggressively than an optimal one, since it cannot rebalance earlier matches when a closer control is needed later.

Taken together, these results suggest that the choice of matching ratio should be guided primarily by the bias-variance tradeoff. In the present setting, fixed-ratio matching dominates variable-ratio matching at every $K \geq 2$: it achieves essentially identical standard errors while incurring substantially less bias, and consequently a lower mean squared error. Among the fixed-ratio configurations, $K=2$ emerges as the preferred operating point, attaining the lowest MSE of any configuration tested (0.000258) by combining a meaningful reduction in standard error relative to $K=1$ (0.1745 vs. 0.2043) with a negligible increase in bias (0.0051 vs. 0.0034). These findings indicate that when an optimal fixed-ratio assignment is feasible, it can be preferable to its greedy variable-ratio counterpart in dynamic 1:k matching contexts.

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